

Edge AI Based Dehumidification System

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To

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ABSTRACT

This project presents an advanced intelligent dehumidification system that integrates embedded edge computing, adaptive power control, and data-driven decision-making to achieve efficient and autonomous humidity regulation. The system employs a DHT22 sensor for real-time measurement of ambient humidity and temperature, along with additional thermal sensing of the Peltier module to monitor operational efficiency. An Arduino-based microcontroller is used for real-time sensing and actuator control, while machine learning inference is performed on an external host system (laptop) using Python. Sensor data is transmitted from the Arduino to the host system via serial communication, where a trained machine learning model predicts optimal control actions. These predictions are then sent back to the Arduino, enabling adaptive and data-driven control of the dehumidification system.

Unlike conventional systems that rely on discrete ON/OFF or fixed multi-level switching, the proposed design implements continuous power modulation of the Peltier module using PWM-driven MOSFET control. This allows fine-grained adjustment of cooling intensity, resulting in improved energy efficiency, reduced thermal stress, and smoother system response. A machine learning model trained on real-world environmental data predicts optimal operating conditions by considering humidity, ambient temperature, and thermal feedback from the system.

The architecture incorporates a closed-loop control mechanism in which sensor data is continuously processed to dynamically regulate actuator behavior. Additionally, the system supports IoT-based monitoring through a web-enabled dashboard, providing real-time visualization of environmental parameters, system performance, and power consumption.

The proposed solution demonstrates a scalable and energy-aware approach to intelligent environmental control, combining edge AI, efficient power electronics, and real-time monitoring. It is suitable for compact, low-cost deployments and can be extended to smart indoor climate management, industrial storage environments, and adaptive IoT-based control system.

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1.1 Overview and Issues Involved

Humidity control is an essential requirement in residential, industrial, and storage environments, as excessive moisture can lead to corrosion of electronic components, mold growth, material degradation, and reduced operational efficiency. Compact dehumidification systems based on thermoelectric (Peltier) modules offer a low-cost and portable solution; however, their performance is often limited by low efficiency, poor thermal management, and lack of intelligent control.

Conventional dehumidifiers typically rely on fixed threshold-based or ON/OFF control strategies, which are not suitable for dynamically changing environmental conditions. These systems often result in inefficient energy usage, frequent switching, and unstable humidity levels. Additionally, thermoelectric systems alone may not provide sufficient moisture removal under varying load conditions.

To address these limitations, modern systems are increasingly integrating data-driven control, hybrid dehumidification techniques, and real-time feedback mechanisms. In this project, a combination of active cooling (Peltier modules) and passive moisture absorption (silica gel desiccant) is used to enhance overall performance. The system also incorporates intelligent control using machine learning, enabling adaptive response to environmental changes.

Several key issues are involved in designing such a system:

- **Low efficiency of thermoelectric modules:** Peltier devices consume significant power and require effective heat dissipation.
- **Dynamic environmental variations:** Humidity and temperature continuously change due to external conditions, requiring adaptive control.
- **Energy consumption:** Continuous operation at full power leads to unnecessary energy usage.
- **Lack of intelligent control:** Traditional systems cannot adapt to changing conditions effectively.

- **Desiccant saturation:** Silica gel loses effectiveness over time and requires regeneration.
- **Need for integrated control:** Both humidity removal and humidity sources must be managed for optimal performance.

These challenges highlight the need for a smart, adaptive, and energy-efficient dehumidification system.

1.2 Problem Statement

Maintaining optimal humidity levels in enclosed environments is a complex task due to continuously changing environmental conditions and the limitations of conventional dehumidification techniques. Traditional systems based on thermoelectric modules or desiccant materials typically operate using simple threshold-based control, leading to inefficient energy usage, poor adaptability, and inconsistent performance.

Thermoelectric dehumidifiers suffer from low efficiency and require precise power regulation to operate effectively. On the other hand, desiccant materials such as silica gel provide passive moisture absorption but require periodic regeneration to maintain functionality. Existing systems rarely integrate both techniques in a coordinated manner.

Furthermore, most compact systems lack intelligent decision-making capabilities and do not incorporate real-time feedback from environmental conditions. The absence of adaptive control results in overcooling, unnecessary power consumption, and reduced system lifespan. Additionally, humidity sources within the environment are not actively controlled, which can further degrade system performance.

Therefore, the problem addressed in this project is the development of an intelligent, hybrid dehumidification system that:

1. Effectively regulates humidity under dynamic environmental conditions
2. Combines active cooling and passive moisture absorption
3. Optimizes energy consumption through adaptive control
4. Incorporates regeneration of desiccant material
5. Controls both humidity removal and humidity sources

6. Utilizes machine learning for real-time decision-making

The challenge lies in integrating these functionalities into a compact, low-cost, and efficient system.

1.3 Proposed Solution

To overcome the limitations of conventional systems, this project proposes an intelligent hybrid dehumidification system that integrates thermoelectric cooling, desiccant-based moisture absorption, and machine learning-based control.

The system uses two Peltier modules for active cooling and condensation, with power controlled using PWM-driven MOSFET modules for efficient and continuous regulation. A DHT22 sensor is used to measure real-time humidity and temperature, and the data is transmitted to a host system where a machine learning model predicts the optimal control level. The predicted output is sent back to the Arduino, which adjusts actuator behavior accordingly.

In addition to active cooling, silica gel beads are incorporated as a passive dehumidification mechanism to enhance moisture absorption. To ensure long-term usability, a PTC heater is used to regenerate the silica gel by removing absorbed moisture. This enables continuous and sustainable operation of the desiccant unit.

A relay-controlled water pump is also integrated into the system to simulate and regulate humidity sources. The pump is automatically switched OFF when humidity exceeds a predefined threshold, preventing further moisture addition and enabling complete environmental control.

The system operates as a closed-loop feedback system where sensor data continuously influences control decisions. By combining machine learning, hybrid dehumidification techniques, and intelligent actuation, the proposed solution achieves improved efficiency, adaptability, and stability compared to traditional systems.

2.1 Existing Solutions

Humidity control systems are widely used in residential, industrial, and environmental applications to maintain air quality and prevent damage caused by excess moisture. Conventional dehumidification technologies primarily include vapor-compression systems and desiccant-based systems. Vapor-compression dehumidifiers operate by cooling moist air below its dew point using a refrigeration cycle, causing water vapor to condense on evaporator coils. This method provides high moisture removal efficiency; however, it suffers from reduced performance at low ambient temperatures, high energy consumption, and the need for refrigerants that may have environmental impacts [11]. Additionally, overcooling of air during condensation often requires reheating, further reducing system efficiency [11].

Desiccant-based dehumidification systems use hygroscopic materials such as silica gel or lithium chloride to absorb moisture directly from the air. These systems operate effectively in low-temperature and high-humidity conditions where condensation-based methods are less efficient [12]. However, they require periodic regeneration of the desiccant material using heat, leading to increased energy consumption and system complexity [12].

In recent years, thermoelectric (Peltier-based) dehumidifiers have emerged as a compact and environmentally friendly alternative. These systems utilize the Peltier effect, where an electric current passing through semiconductor junctions creates a temperature difference, enabling one side of the module to cool while the other heats up. Moist air passing over the cold surface condenses into liquid water, which is then collected [2], [3]. Thermoelectric systems are advantageous due to their compact size, absence of refrigerants, low noise, and high reliability in small-scale applications [3].

Despite these benefits, thermoelectric dehumidifiers exhibit significant limitations. Their coefficient of performance (COP) is generally low due to Joule heating and thermal losses, and their moisture removal capacity is considerably lower than that of conventional

vapor-compression systems [4], [6]. As a result, their application is typically limited to small or localized environments.

To improve performance, various studies have investigated the impact of operational and design parameters on thermoelectric systems. Factors such as airflow rate, input current, and heat sink design have been shown to significantly influence system efficiency and moisture removal rate [6]. Increasing airflow improves heat transfer and condensation rates up to an optimal point, beyond which additional fan power reduces overall efficiency [6]. Similarly, optimizing heat sink geometry and reducing thermal resistance enhances cooling performance [2]. Multi-stage thermoelectric configurations have also been explored, where multiple modules are used to achieve greater temperature differences and improved condensation efficiency; however, these systems introduce higher complexity and power consumption [3].

In terms of control strategies, traditional systems rely on threshold-based ON/OFF control, where operation is triggered when humidity exceeds a predefined limit. While simple to implement, this approach leads to frequent switching, oscillations, and inefficient energy usage under dynamic environmental conditions [11]. To address these limitations, advanced control techniques such as fuzzy logic and PID controllers have been proposed. Fuzzy controllers improve performance under measurement uncertainty and environmental variability by using rule-based decision-making [7]. Intelligent air conditioning and humidity control systems have also been implemented using fuzzy logic and PID tuning to achieve smoother response and improved energy efficiency [7], [8].

More recently, machine learning techniques have been applied to HVAC and environmental control systems. Supervised learning models and neural networks can analyze complex relationships between environmental parameters and system performance, enabling predictive and adaptive control [9]. Studies have shown that AI-driven optimization can significantly reduce energy consumption while maintaining desired environmental conditions [10]. These approaches demonstrate the potential of data-driven control strategies for improving the efficiency and adaptability of dehumidification systems.

2.2 Methodology

The methodology designed for this project follows a structured framework encompassing

requirement analysis, literature review, real-world data acquisition, architectural design, machine learning model development, embedded implementation, system integration, and performance evaluation. The system is developed as a closed-loop intelligent control framework that integrates sensing, edge computation, and adaptive actuation for efficient humidity regulation.

2.2.1 Requirement Analysis

The system is required to maintain target humidity levels under dynamically changing environmental conditions while ensuring energy-efficient operation. Functional requirements include real-time sensing of humidity, ambient temperature, and internal thermal parameters, embedded decision-making, adaptive power control, and real-time monitoring.

Non-functional requirements include low cost, compact design, reliability, modularity, scalability, and minimal power consumption. The system must also operate autonomously without dependence on external computation and support wireless monitoring for user accessibility.

2.2.2 Literature Review and Theoretical Foundation

Relevant literature on thermoelectric dehumidification [3], [4], [5], thermal optimization techniques, intelligent control strategies such as fuzzy logic and PID [6], [7], [8], and machine learning-based HVAC optimization [9], [10], [11], [12] was analyzed to understand existing approaches and identify limitations.

The review highlights that while thermoelectric systems offer compact and environmentally friendly solutions, their performance is highly dependent on thermal management and control strategies. Additionally, existing implementations largely rely on static or rule-based control mechanisms, which lack adaptability. These observations motivate the development of an embedded, data-driven control system capable of performing real-time inference and adaptive decision-making directly on hardware.

2.2.3 Architecture Design

The system is designed as a multi-layer closed-loop architecture consisting of the following components:

- **Sensing Layer:**

Digital sensors measure ambient humidity, ambient temperature, and internal thermal conditions of the Peltier module.

- **Embedded Control Layer:**

The Arduino microcontroller is responsible for sensor data acquisition and actuator control. Machine learning inference is performed on a host system (laptop), which receives sensor data via serial communication, processes it using a trained model, and transmits control commands back to the Arduino. This distributed architecture enables real-time adaptive control while overcoming computational limitations of embedded hardware.

- **Machine Learning Layer**

A trained model predicts optimal control signals based on environmental and thermal inputs.

- **Actuation Layer:**

The Peltier module and cooling fans are controlled using PWM-based power modulation through MOSFET drivers, enabling continuous control.

- **Monitoring Layer:**

Wireless communication enables real-time data transmission to a web-based dashboard for visualization and analysis.

Data flows cyclically as:

Environment → Sensors → Arduino Processing → ML Inference → PWM Control → Actuators → Environment

This continuous feedback loop ensures adaptive and stable humidity regulation.

2.2.4 Dataset Definition and Feature Engineering

A real-world dataset is constructed by collecting environmental and system-level parameters during operation. The dataset includes humidity, ambient temperature, cold-side temperature, and hot-side temperature of the Peltier module.

Derived features such as humidity deviation from the target range and temperature gradients are computed to enhance model performance. The feature vector is defined as:

$$X = [\text{humidity, ambient temperature, cold-side temperature, hot-side temperature}]$$

The output represents the optimal control signal, corresponding to the required power level for the Peltier module. This approach enables the model to capture both environmental conditions and internal system behavior.

2.2.5 Machine Learning Model Training

A lightweight machine learning model is selected to ensure efficient embedded deployment. The model is trained using the collected dataset, with preprocessing steps including normalization, feature scaling, and dataset splitting into training and testing sets.

The model learns the relationship between environmental conditions and optimal control actions, enabling predictive and adaptive decision-making. Training is performed using standard optimization techniques to minimize prediction error.

The trained model is then converted into a format suitable for deployment on the embedded system, enabling real-time inference with minimal computational overhead.

The trained machine learning model is deployed on a host system (laptop) rather than directly on the embedded controller. Real-time sensor data is transmitted from the Arduino to the host system using serial communication. The model performs inference on incoming

data and sends the predicted control signal back to the Arduino. This approach ensures efficient computation while maintaining real-time system responsiveness.

2.2.6 Firmware and Software Integration

The embedded firmware running on the is responsible for sensor data acquisition, execution of the machine learning model, and generation of PWM control signals. Sensor readings are continuously collected and processed to determine optimal operating conditions.

The system dynamically adjusts the power supplied to the Peltier module and cooling fans using PWM signals, ensuring smooth and efficient operation.

On the software side, a web-based dashboard is implemented to provide real-time visualization of humidity, temperature, and system performance. Wireless communication protocols are used to transmit data between the embedded device and the dashboard, enabling remote monitoring and user interaction.

2.2.7 Full System Validation

The integrated system is tested under varying environmental conditions to evaluate performance, stability, and efficiency. Controlled experiments are conducted by introducing changes in humidity and observing system response.

Performance metrics include response time, accuracy of control decisions, energy consumption, and stability of operation. The system demonstrates effective humidity regulation, smooth adaptive control, and efficient energy usage under different conditions.

The results confirm the robustness and practical applicability of the proposed intelligent dehumidification system.

3.1 Software Requirements

The software component of the system is responsible for data acquisition, machine learning-based decision making, and real-time control of actuators. The system uses a combination of embedded programming and high-level data processing tools.

The Arduino Integrated Development Environment (IDE) is used for programming the microcontroller, enabling sensor interfacing, serial communication, and actuator control through PWM and relay signals. The microcontroller continuously reads environmental parameters such as humidity and temperature and transmits them to an external host system.

Python is used as the primary platform for implementing the machine learning model. The model is developed using libraries such as NumPy and Scikit-learn, which facilitate efficient data handling and training of the classification model. The trained model performs real-time inference by processing incoming sensor data and predicting the appropriate control level.

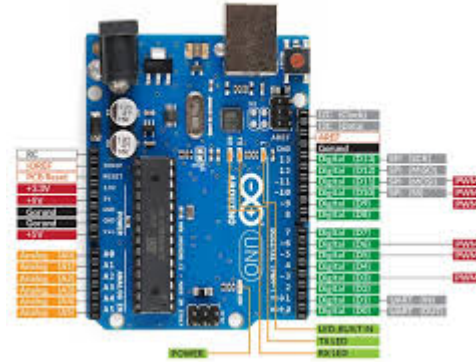
Serial communication is established between the Arduino and the host system, enabling bidirectional data exchange. The Arduino sends sensor readings, while the Python program returns control signals in the form of discrete levels, which are then used to adjust system behavior.

The software architecture is designed to be modular and efficient, allowing real-time operation while maintaining low computational overhead on the microcontroller.

3.2 Hardware Requirements

The hardware design consists of sensing units, control units, and actuation systems, all integrated to form a complete dehumidification system.

3.2.1 Arduino Uno



(Figure 3.1)

The Arduino Uno serves as the central control unit of the system. It interfaces with sensors, communicates with the machine learning model via serial communication, and controls actuators such as the Peltier modules and relay module. Its simplicity, reliability, and ease of programming make it suitable for real-time embedded applications.

3.2.2 DHT22 Sensor

The DHT22 sensor is used to measure environmental humidity and temperature. It provides accurate and stable readings, which are essential for reliable machine learning predictions and system control.



(Figure 3.2)

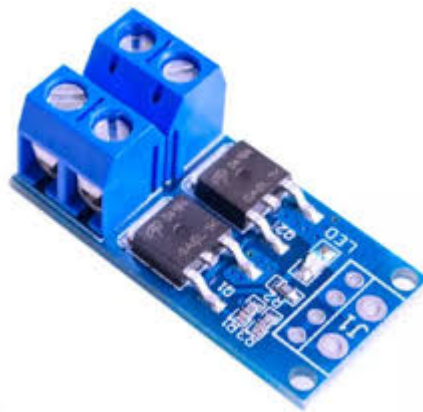
3.2.3 Peltier Modules (TEC)



(Figure 3.3)

Two Peltier modules are used for active cooling and condensation. When powered, they create a temperature difference that causes moisture in the air to condense, thereby reducing humidity. Using two modules improves cooling capacity and enhances overall dehumidification performance.

3.2.4 MOSFET Driver Modules (XY-MOS)



(Figure 3.4)

Two XY-MOS modules are used to control the power supplied to the Peltier devices. These modules enable PWM-based control, allowing the system to regulate cooling intensity efficiently. This prevents unnecessary power consumption and provides smooth control

compared to simple ON/OFF switching.

3.2.5 Silica Gel Desiccant



(Figure 3.5)

Silica gel beads are used as a passive dehumidification component. They absorb moisture directly from the air, supplementing the cooling-based dehumidification process. This hybrid approach improves system efficiency and performance, especially under low cooling conditions.

3.2.6 PTC Heater



(Figure 3.6)

A PTC (Positive Temperature Coefficient) heater is used to regenerate the silica gel. Over time, the desiccant becomes saturated with moisture and loses effectiveness. The PTC heater provides controlled heating to remove absorbed moisture, restoring the silica gel for reuse. Its self-regulating nature ensures safe operation.

3.2.7 Relay Module



(Figure 3.7)

A 4-channel relay module is used, with one channel dedicated to controlling a water pump. The relay allows safe switching of higher power devices using low-power control signals from the Arduino. The pump is used to simulate or control humidity input and is turned OFF when humidity exceeds the desired level.

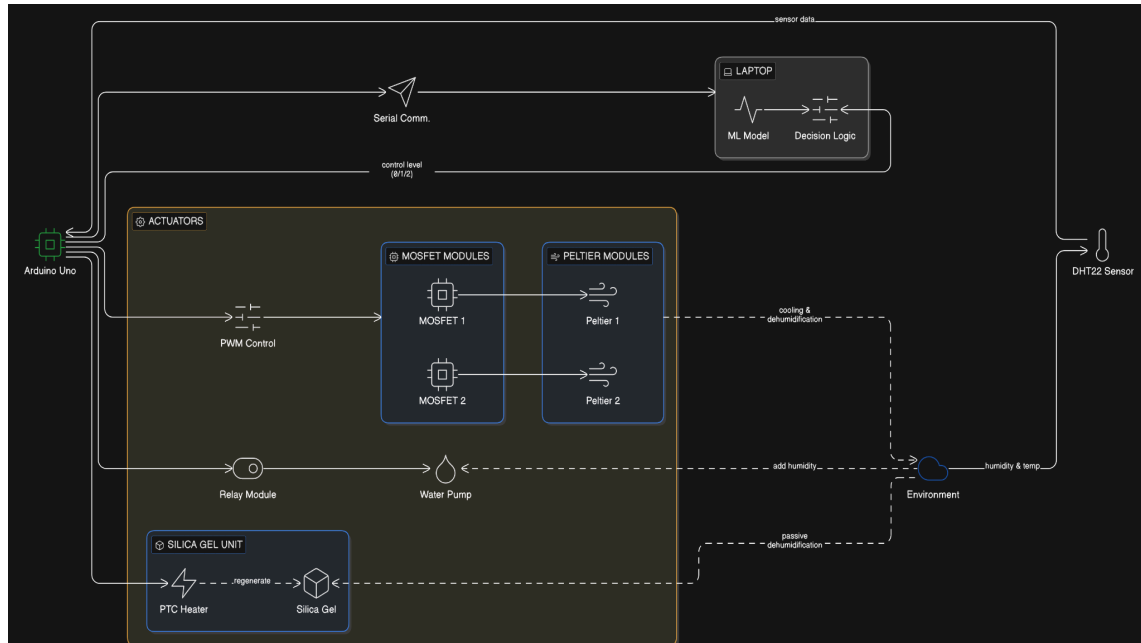
3.2.8 Power Supply



(Figure 3.8)

A 12V, 5A power supply is used to power the Peltier modules and other high-power components. The Arduino is powered separately through USB or regulated supply. Proper grounding is maintained across all components to ensure stable and safe operation.

3.3 Block Diagram



(Figure 3.9)

The block diagram of the system represents the interaction between sensing, processing, and actuation units.

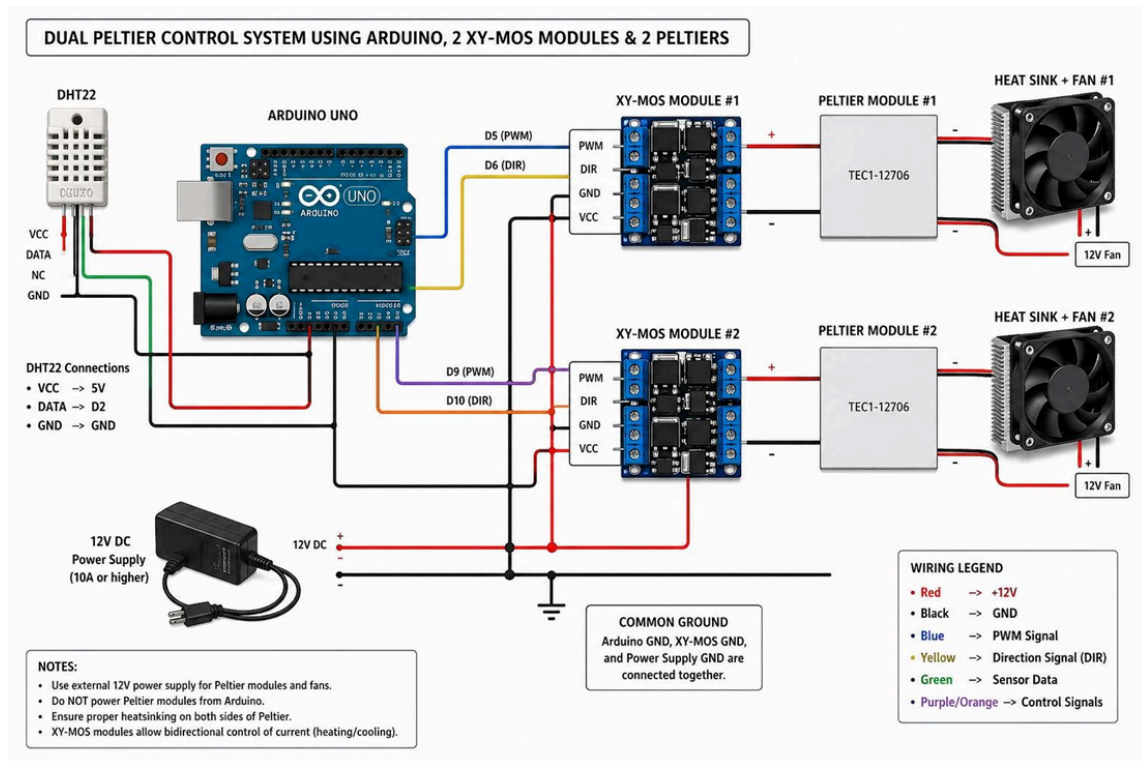
The DHT22 sensor measures humidity and temperature and sends this data to the Arduino microcontroller. The Arduino transmits the sensor data to a host system via serial communication. The machine learning model processes this data and predicts the appropriate control level.

The predicted control signal is sent back to the Arduino, which adjusts the PWM signals controlling the MOSFET driver modules. These drivers regulate the power supplied to the Peltier modules, controlling the cooling intensity.

Simultaneously, the relay module controls the water pump based on the predicted control level, ensuring that humidity input is regulated. The silica gel absorbs moisture passively, while the PTC heater regenerates it when required.

This forms a closed-loop system where environmental conditions continuously influence system behavior.

3.4 Circuit Diagram



(Figure 3.10)

The circuit diagram illustrates the electrical connections between all components of the system.

The DHT22 sensor is connected to the Arduino through a digital input pin, providing humidity and temperature readings. The PWM output pins of the Arduino are connected to the control inputs of the XY-MOS modules, which in turn drive the Peltier modules using an external power supply.

The relay module is connected to a digital output pin of the Arduino and is used to control the water pump. The pump is powered separately, and the relay ensures electrical isolation between the control and power circuits.

The PTC heater is connected to the power supply through appropriate switching and operates independently for desiccant regeneration.

The implementation phase represents the practical realization of the intelligent dehumidification system using a hybrid architecture consisting of an Arduino microcontroller, external machine learning processing on a host system (laptop), and MOSFET-based actuation. Unlike the earlier design, the system does not perform embedded inference on a microcontroller; instead, it utilizes serial communication to enable real-time decision-making using a trained machine learning model. This chapter presents the circuit implementation, signal flow, testing procedures, and performance evaluation of the system.

4.1 Circuit Description and Signal Flow

The system consists of sensing, processing, communication, and actuation subsystems working together in a closed-loop configuration.

1. Sensor Subsystem

The sensing unit consists of a DHT22 sensor used to measure ambient humidity and temperature. The sensor is interfaced with the Arduino using a digital communication pin.

Signal Flow:

Environment → DHT22 Sensor → Arduino

2. Data Acquisition and Communication (Arduino)

The Arduino Uno acts as the data acquisition unit. It reads sensor values and transmits humidity and temperature data to the host system via serial communication (USB).

Signal Flow:

Sensor Data → Arduino → Serial (USB) → Laptop

3. Machine Learning Processing (Laptop)

The machine learning model is implemented on a laptop using Python. A supervised classification model (multinomial logistic regression) is trained on environmental data to predict optimal control levels.

The laptop receives real-time sensor data from Arduino, processes it using the trained model, and generates a control output.

Signal Flow:

Arduino Data → Laptop → ML Model → Control Prediction

4. Control Signal Generation (Arduino)

The predicted control signal is sent back to the Arduino via serial communication. Based on this input, the Arduino generates PWM signals to control actuator power levels.

Signal Flow:

Laptop → Arduino → PWM Output

5. Switching Subsystem (XY-MOS Modules)

Two XY-MOS modules are used as high-power MOSFET drivers. These modules receive PWM signals from the Arduino and regulate the current supplied to the Peltier modules.

This allows smooth control of power instead of simple ON/OFF switching.

6. Actuator Subsystem (Peltiers and Fans)

The actuator unit consists of:

- Two Peltier modules for cooling and moisture condensation
- Cooling fans for heat dissipation

The Peltiers create a temperature gradient, causing moisture in the air to condense, thus

reducing humidity.

Signal Flow:

PWM → XY-MOS → Peltiers → Cooling → Dehumidification

7. Complete System Feedback Loop

The system operates continuously as a closed-loop:

- Sensors measure environment
- Arduino sends data to laptop
- ML model predicts control level
- Arduino adjusts PWM
- Peltiers change humidity
- Sensors detect updated conditions

This loop enables adaptive and intelligent humidity control.

4.2 Testing Procedure

The system is tested in multiple stages to ensure proper operation.

1. Hardware Testing

- Sensor readings verified for accuracy
- XY-MOS modules tested using fan loads
- Peltiers tested for heating/cooling behavior
- Power supply stability verified

2. Communication Testing

- Serial communication between Arduino and laptop verified
- Data transmission checked for correctness
- Control signals successfully received back

3. Machine Learning Validation

- Model tested using generated dataset

- Predictions verified for different humidity levels
- Control output changes smoothly across conditions

4. Integrated System Testing

- Full system assembled and tested
- Humidity changes introduced manually
- System response observed

As humidity increases → PWM increases As humidity decreases → PWM reduces

5. Stability Testing

- Continuous operation tested
- No overheating or instability observed
- System maintains consistent behavior

4.3 Results and Discussion

The system demonstrates effective and adaptive humidity control.

1. Sensor Performance

Sensors provide stable and consistent readings suitable for real-time control.

2. Adaptive Control Behavior

The ML model enables smooth control instead of abrupt switching, improving system stability.

3. Power Control Efficiency

XY-MOS modules allow efficient PWM-based regulation of Peltier modules.

4. Dehumidification Performance

- High humidity → increased cooling
- Low humidity → reduced power

- System adapts continuously

5. System Reliability

The system operates reliably over extended periods without failure.

6. Overall Performance

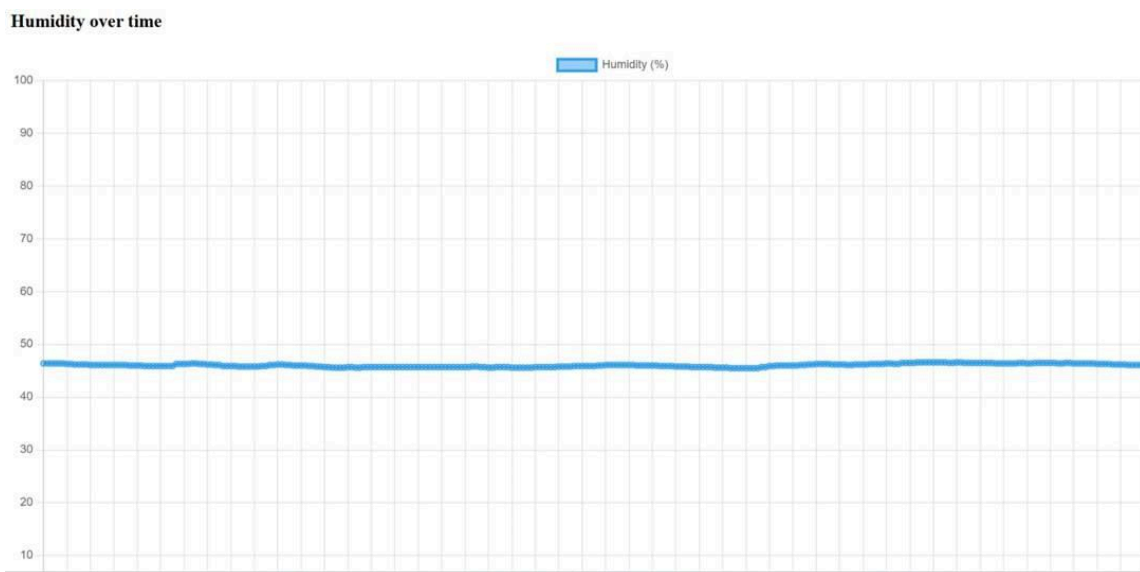
The hybrid architecture successfully combines:

- Real-time sensing
- Data-driven decision making
- Efficient power control

This results in improved adaptability and energy efficiency compared to conventional threshold-based systems.

4.3.1 Comparative analysis of the system, with AI, and without AI:

To evaluate the effectiveness of the proposed machine learning-based control system, a comparative analysis was performed between conventional control (without AI) and the proposed AI-based system.



(Figure 4.1 Single Peltier Module Without AI based PWM)



(Figure 4.2 Working of System without AI)

Figure 4.2 shows the system response using traditional threshold-based control. It can be observed that the system exhibits abrupt switching behavior, leading to fluctuations and less stable humidity regulation.



(Figure 4.3 Working of System with AI)

Figure 4.3 illustrates the performance of the AI-based control system. The response is smoother and more adaptive, with gradual changes in control output based on environmental conditions.

The comparison highlights the following improvements:

- Reduced oscillations in humidity levels
- Smoother control of actuator power
- Improved stability and responsiveness
- Better energy efficiency due to adaptive control

These results demonstrate that the machine learning-based approach significantly enhances system performance compared to conventional methods.

4.3.2 Power Consumption Analysis:

To evaluate the energy efficiency of the proposed system, a comparative analysis of power consumption was conducted for both conventional control (without machine learning) and the proposed ML-based control system.



(Figure 4.4 Without AI based PWM)

Figure 4.4 illustrates the power consumption pattern of the system operating without machine learning. It can be observed that the system follows an abrupt ON/OFF switching behavior, resulting in sharp fluctuations between maximum and zero power levels. This leads to inefficient energy usage and increased system instability.



(Figure 4.5 AI based PWM)

Figure 4.5 shows the power consumption of the system using the ML-based control approach. Unlike the conventional method, the ML model adjusts the control signal gradually using PWM, resulting in smooth variations in power consumption. The system reduces power as humidity decreases, thereby optimizing energy usage.

The comparison clearly demonstrates that the ML-based system:

- Reduces average power consumption
- Eliminates unnecessary ON/OFF switching
- Provides smoother and more stable control
- Improves overall energy efficiency

Thus, the machine learning approach enhances both system performance and power efficiency.

Chapter 5

Conclusion

The development of an intelligent, edge-enabled dehumidification system represents a significant advancement in integrating embedded systems with modern data-driven control techniques. The primary objective of this project was to design a compact, energy-efficient, and adaptive humidity control system capable of operating autonomously using real-time environmental feedback and embedded machine learning. The successful implementation of this system demonstrates the feasibility of combining edge computing, efficient power electronics, and intelligent control within a unified architecture.

At the core of the system lies the Arduino UNO microcontroller, which enables real-time sensing, host device machine learning inference, and wireless communication. By eliminating dependence on external computation, the system achieves true standalone operation with reduced latency and improved reliability. The integration of multiple sensors, including ambient humidity, temperature, and internal thermal measurements of the Peltier module, provides a comprehensive understanding of both environmental conditions and system performance.

A key innovation of this work is the implementation of continuous power control using PWM-based MOSFET drivers. Unlike traditional discrete switching approaches, this method allows fine-grained modulation of the Peltier module and cooling fans. As a result, the system achieves improved energy efficiency, reduced thermal stress, and smoother control behavior. The ability to dynamically adjust power levels based on real-time conditions significantly enhances overall system performance.

The incorporation of a machine learning model enables intelligent decision-making by capturing complex relationships between environmental variables and optimal control actions. By utilizing real-world data for training, the system demonstrates improved adaptability and accuracy compared to static or rule-based control strategies. The embedded model allows the system to respond proactively to changes in humidity while maintaining stable operation without oscillations.

Another important contribution is the integration of IoT-based monitoring through a web dashboard. The system provides real-time visualization of humidity, temperature, and control signals, enhancing user awareness and enabling remote supervision. This feature improves usability and supports further analysis of system behavior over extended periods.

Experimental validation confirms that the system operates reliably under varying environmental conditions. The system demonstrates accurate sensing, stable control, efficient power utilization, and consistent humidity reduction. The closed-loop architecture ensures continuous adaptation, making the system robust against environmental fluctuations. Long-duration testing further validates the thermal stability and reliability of both hardware and software components.

From an engineering perspective, this project highlights the effectiveness of combining embedded intelligence with efficient hardware design to solve real-world environmental control problems. The system successfully achieves its objectives:

- Accurate real-time sensing of humidity, temperature, and thermal conditions
- Adaptive, machine learning-based control for efficient dehumidification
- Continuous power regulation using PWM for improved energy efficiency
- Autonomous operation through embedded edge computing
- Real-time monitoring and user interaction via IoT dashboard
- Scalable and modular system architecture

The proposed system establishes a strong foundation for future enhancements. Potential improvements include the use of advanced machine learning models such as reinforcement learning for predictive control, integration of additional environmental sensors for

multi-parameter optimization, implementation of cloud-based analytics for long-term data insights, and further optimization of thermal and mechanical design for enhanced efficiency.

In conclusion, this work demonstrates the practical potential of intelligent embedded systems in environmental control applications. By integrating edge AI, adaptive control, and IoT-based monitoring, the system offers a scalable, energy-aware, and responsive solution for modern humidity regulation challenges. It serves as a foundation for future research and development in smart climate control and intelligent IoT systems.

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Appendix

Appendix A: Arduino Code

The Arduino program is responsible for:

- Reading humidity and temperature from DHT22 sensor
- Sending data to host system via serial communication
- Receiving control level from machine learning model
- Generating PWM signals for Peltier control
- Controlling relay for pump operation

```
#include <DHT.h>
```

```
// ===== SENSOR SETUP =====
```

```
#define DHTPIN 2
```

```
#define DHTTYPE DHT22
```

```
DHT dht(DHTPIN, DHTTYPE);
```

```
// ===== PINS =====
```

```
int pwm1 = 5;    // XY-MOS 1 (Peltier 1)
```

```
int pwm2 = 6;    // XY-MOS 2 (Peltier 2)
```

```
int relayPin = 8; // Relay (Pump)
```

```
// ===== SETUP =====
```

```
void setup() {
```

```
    Serial.begin(9600);
```

```
    dht.begin();
```

```
    pinMode(pwm1, OUTPUT);
```

```
    pinMode(pwm2, OUTPUT);
```

```
    pinMode(relayPin, OUTPUT);
```

```
    digitalWrite(relayPin, HIGH); // Pump OFF initially (active LOW)
```

```
}
```

```
// ===== LOOP =====
```

```
void loop() {
```

```

// === READ SENSOR ===
float h = dht.readHumidity();
float t = dht.readTemperature();

// Send only valid readings
if (!isnan(h) && !isnan(t)) {
    Serial.print(h);
    Serial.print(",");
    Serial.println(t);
}

delay(200);

// === RECEIVE ML OUTPUT ===
if (Serial.available()) {
    int level = Serial.parseInt(); // 0,1,2

    int pwmVal1 = 0;
    int pwmVal2 = 0;

    // ===== CONTROL LOGIC =====
    if (level == 0) {
        pwmVal1 = 0;
        pwmVal2 = 0;
        digitalWrite(relayPin, LOW); // Pump ON
    }

    else if (level == 1) {
        pwmVal1 = 150; // Medium
        pwmVal2 = 0;
        digitalWrite(relayPin, LOW); // Pump ON
    }

    else if (level == 2) {
        pwmVal1 = 255; // Full
        pwmVal2 = 255; // Full
        digitalWrite(relayPin, HIGH); // Pump OFF
    }

    // Apply PWM
    analogWrite(pwm1, pwmVal1);
    analogWrite(pwm2, pwmVal2);
}

delay(800);
}

```

Appendix B: Machine Learning Code (Python)

```
import serial
import time
import numpy as np

# ===== MODEL PARAMETERS (YOUR TRAINED VALUES) =====
w = np.array([
    [-1.95906435, 0.13935992],
    [ 0.1653572 , 0.35332168],
    [ 1.79370709, -0.49268199]
])

b = np.array([94.1027715, -8.1825106, -85.92026089])

# ===== PREDICTION FUNCTION =====
def predict(h, t):
    x = np.array([h, t])
    scores = np.dot(w, x) + b
    return np.argmax(scores)

# ===== SERIAL SETUP =====
ser = serial.Serial('COM5', 9600) # CHANGE COM PORT
time.sleep(2)

print("System started...")

# ===== MAIN LOOP =====
while True:
    try:
        line = ser.readline().decode(errors='ignore').strip()

        # Ignore unwanted lines
        if ',' not in line:
            continue

        # Parse sensor data
        h, t = map(float, line.split(','))

        # ML prediction
        level = predict(h, t)

        print(f'H: {h:.2f}, T: {t:.2f} → Level: {level}')

        # Send to Arduino
        ser.write(f'{level}\n'.encode())

    except Exception as e:
        print("Error:", e)
```

Appendix C: Dataset Description

The dataset consists of humidity and temperature values mapped to control levels.

Features:

- Humidity (%)
- Temperature (°C)

Output:

- Control Level (0, 1, 2)

The dataset is generated to simulate real-world environmental variations and used for training the machine learning model.

Appendix D: Additional System Components

Silica Gel:

Used for passive moisture absorption.

PTC Heater:

Used for regeneration of silica gel.

Relay Module:

Used to control water pump based on humidity conditions.

Water Pump:

Used to simulate humidity input and test system response.

Appendix E: Safety Considerations

- Proper isolation between high-power and control circuits
- Use of external power supply for Peltiers
- Common grounding maintained across system
- Avoidance of back current into microcontroller